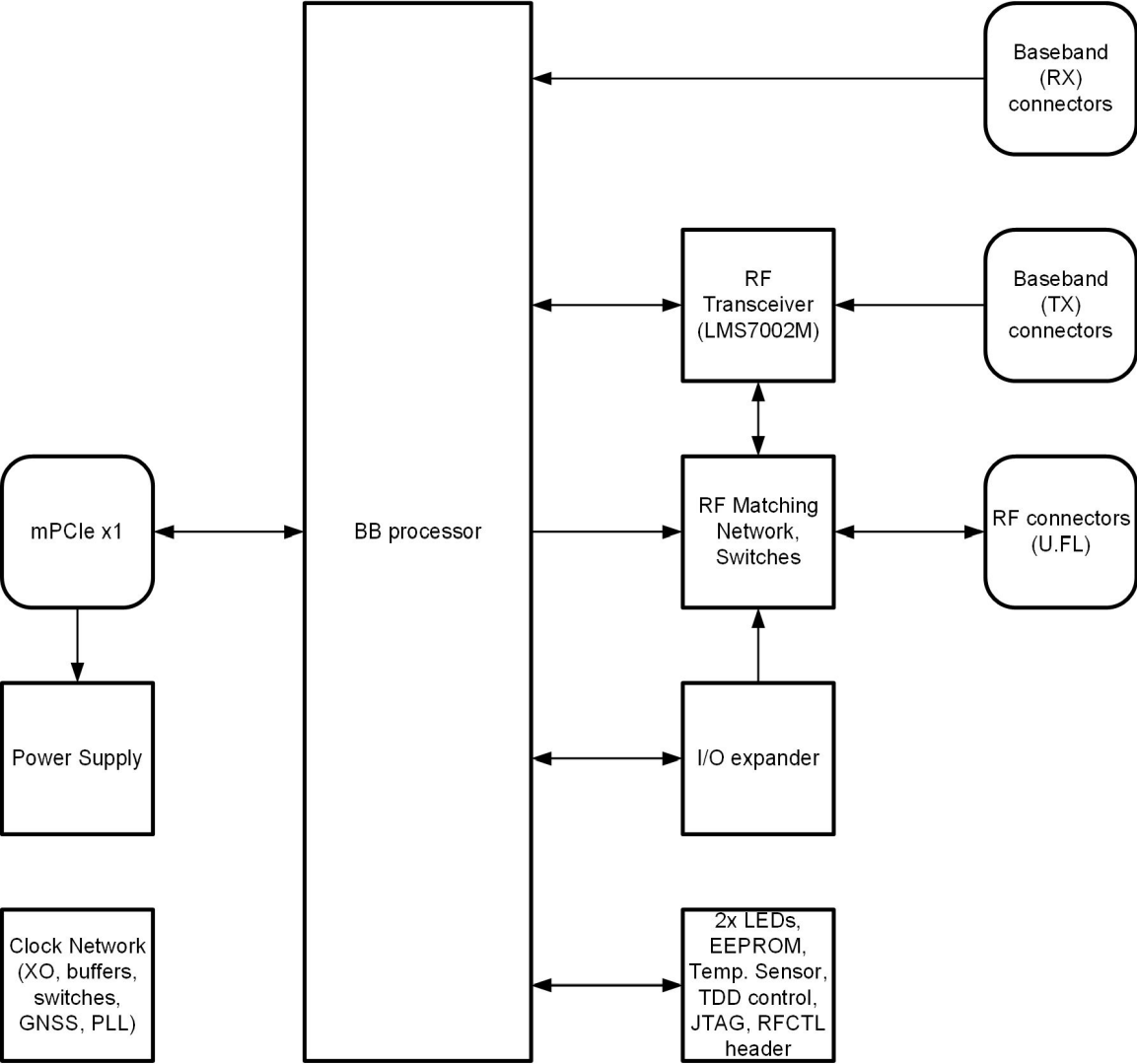
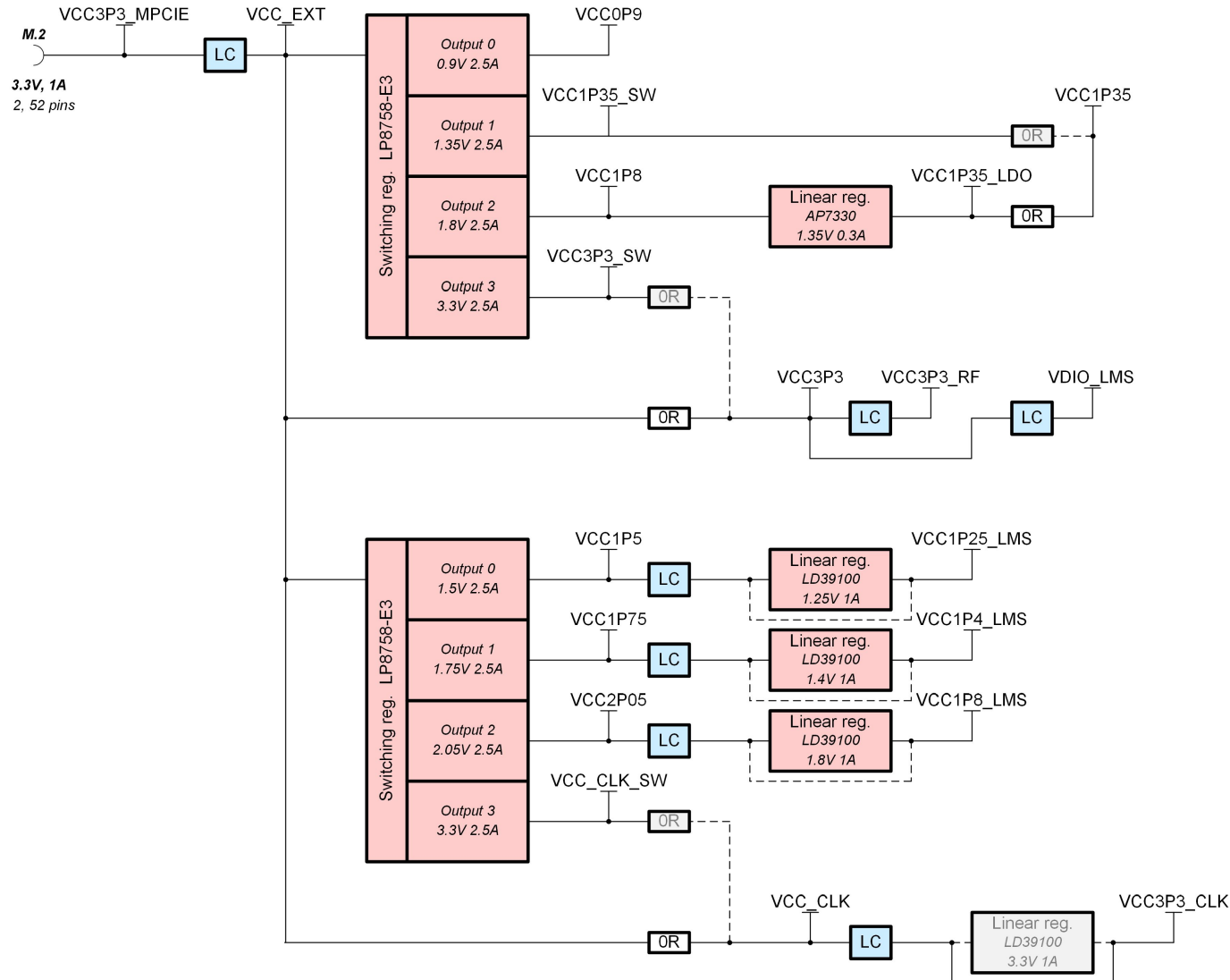


Block diagram



Power diagram



Project name: **LimeSDR-Micro_mPCIe_1v4.PrjPcb**

Title: **Power diagram**

Version: **1.4**

Variant: **Default**

Date: **6/16/2026**

Time: **2:47:59 PM**

Sheet **2** of **11**

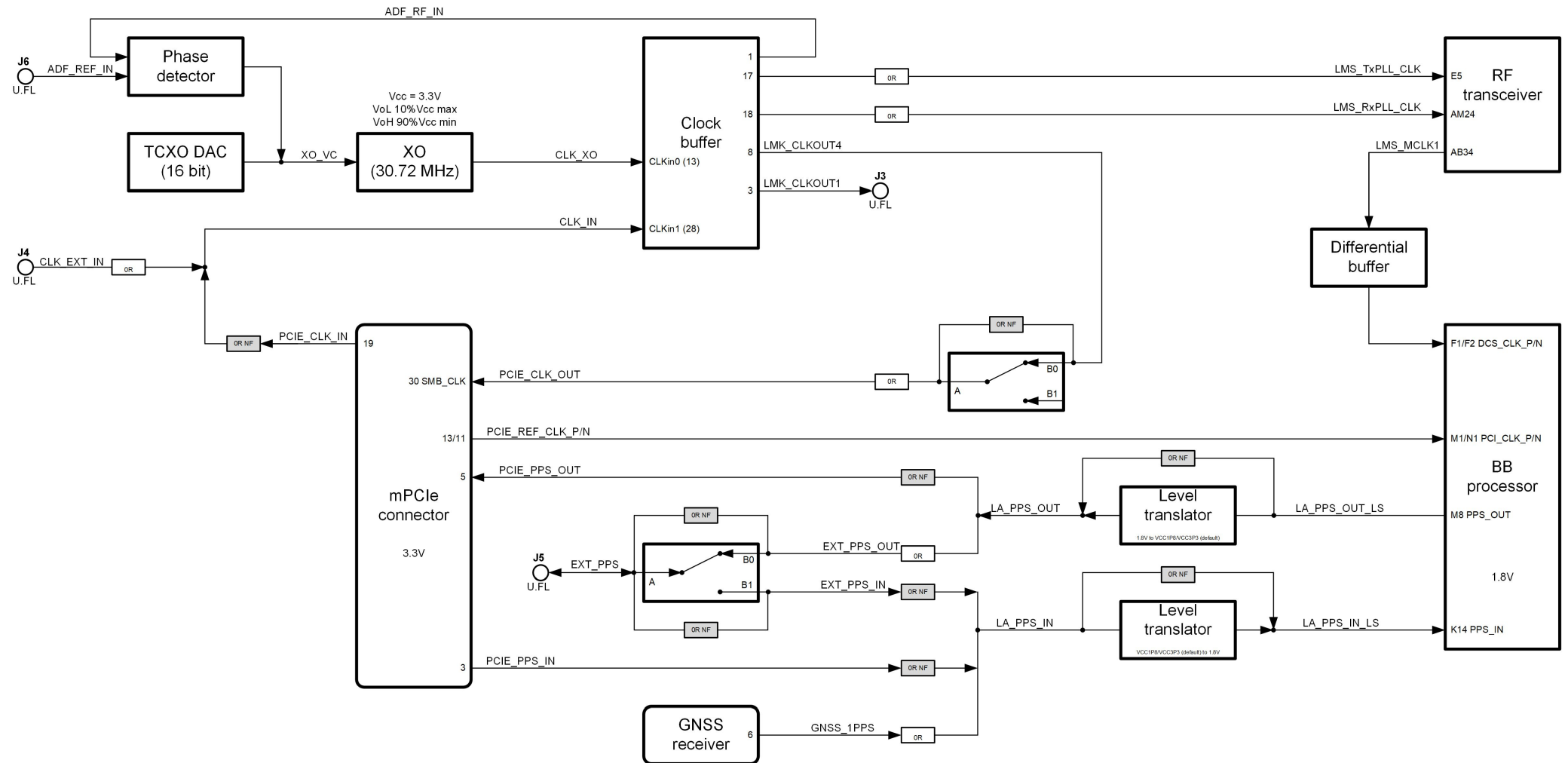
File: **02_Power_Diagram.SchDoc**

Size: **A4**

Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom



Clock diagram



Project name: **LimeSDR-Micro_mPCIe_1v4.PrjPcb**

Title: **Clock diagram**

Version: **1.4**

Variant: **Default**

Date: **6/16/2026**

Time: **2:47:59 PM**

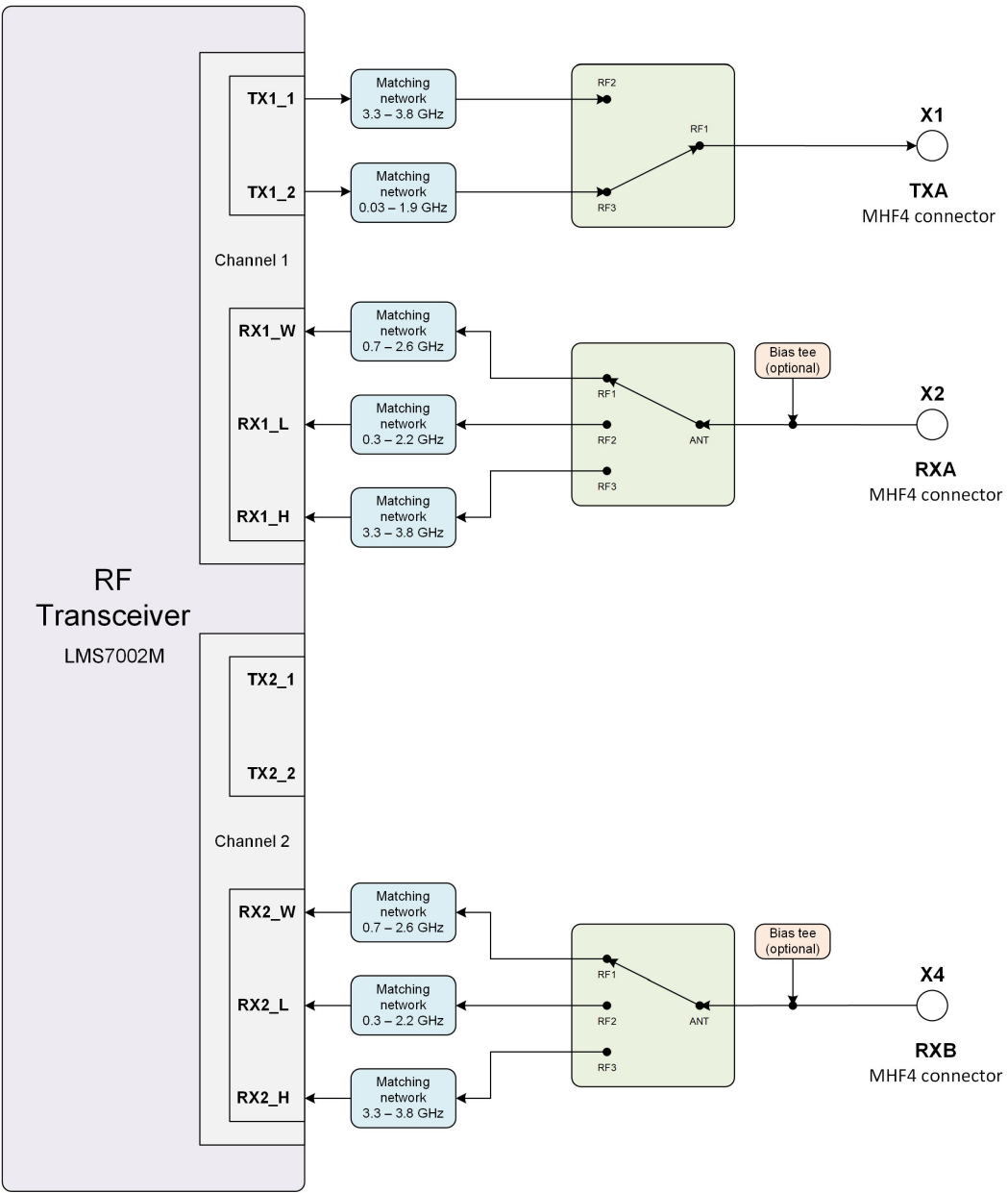
Sheet **3** of **11**

File: **03_Clock_Diagram.SchDoc**

Size: **A4**

Lime Microsystems
Surrey Tech Centre
Guildford GU2 7YG
Surrey
United Kingdom

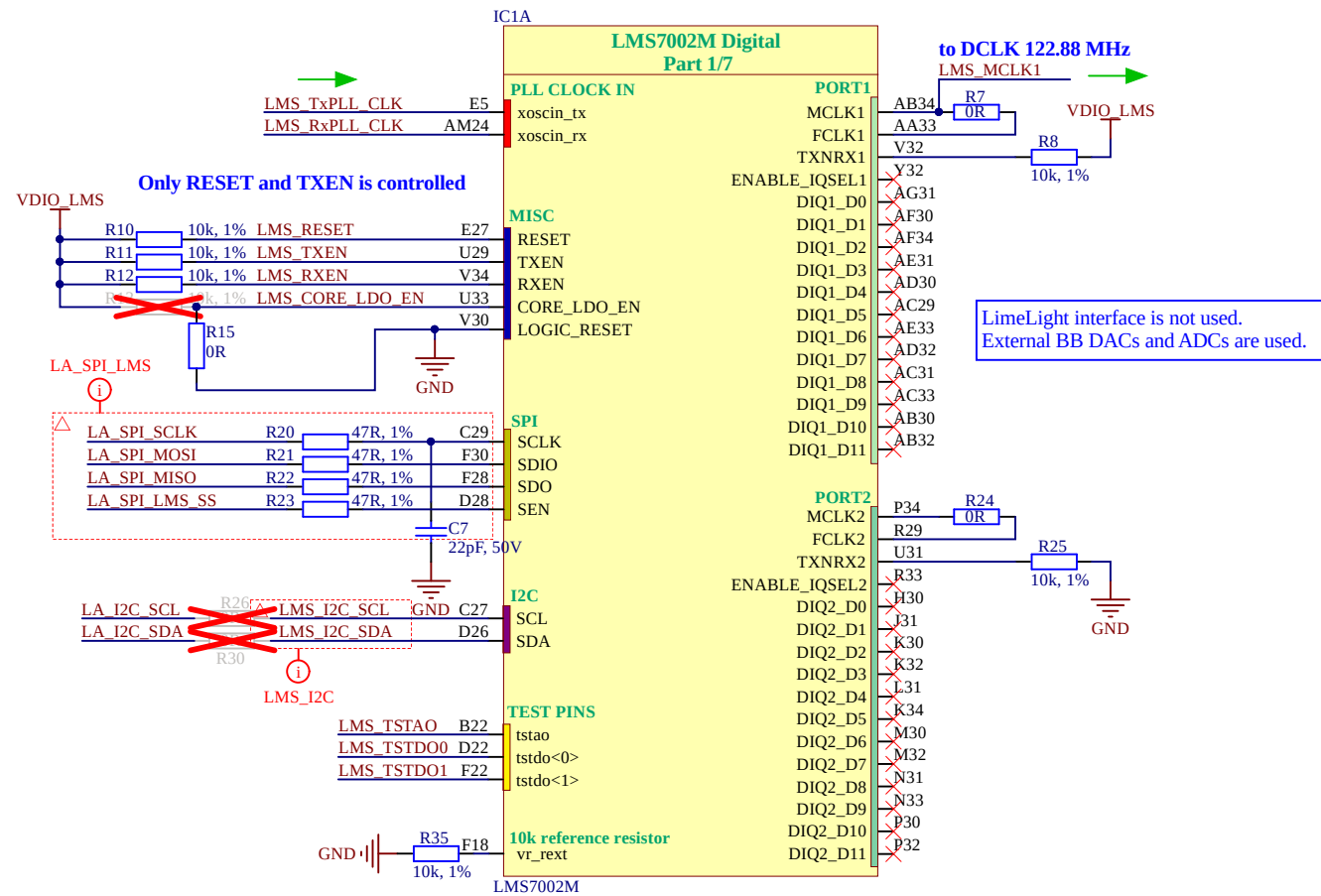




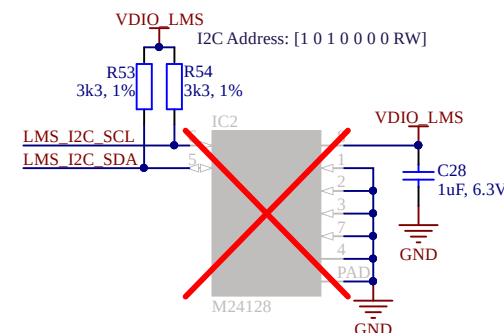
Project name: LimeSDR-Micro_mPCIe_1v4.PrjPcb			
Title: RF diagram		Lime Microsystems Surrey Tech Centre Guildford GU2 7YG Surrey United Kingdom	
Version: 1.4	Variant: Default		
Date: 6/16/2026	Time: 2:48:00 PM		
File: 04_RF_Diagram.SchDoc	Sheet4 of 11	Size: A4	

LMS7002M misc

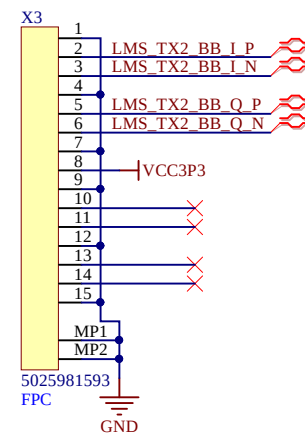
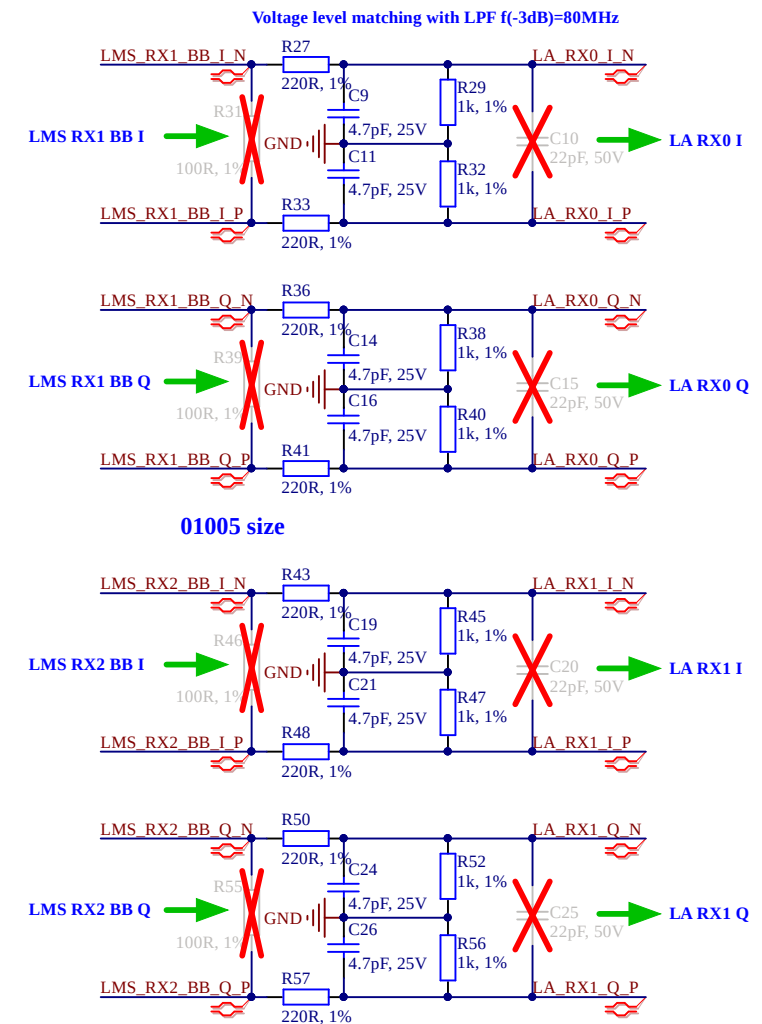
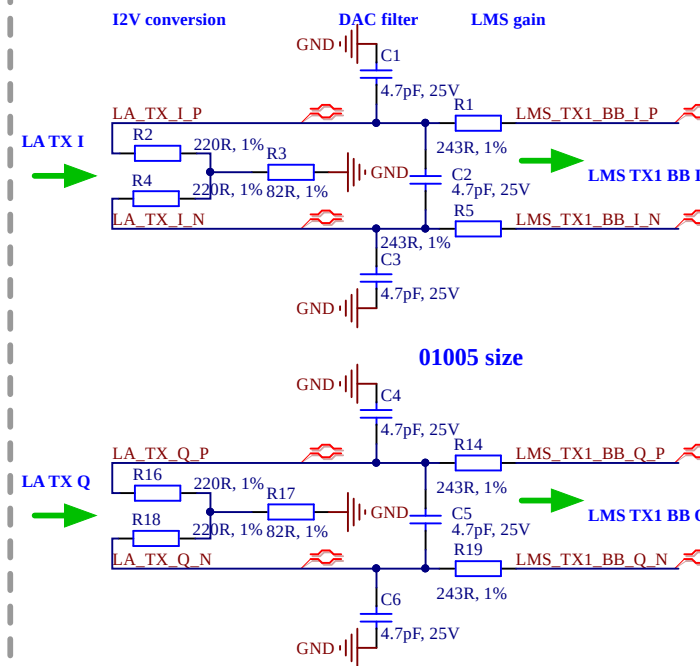
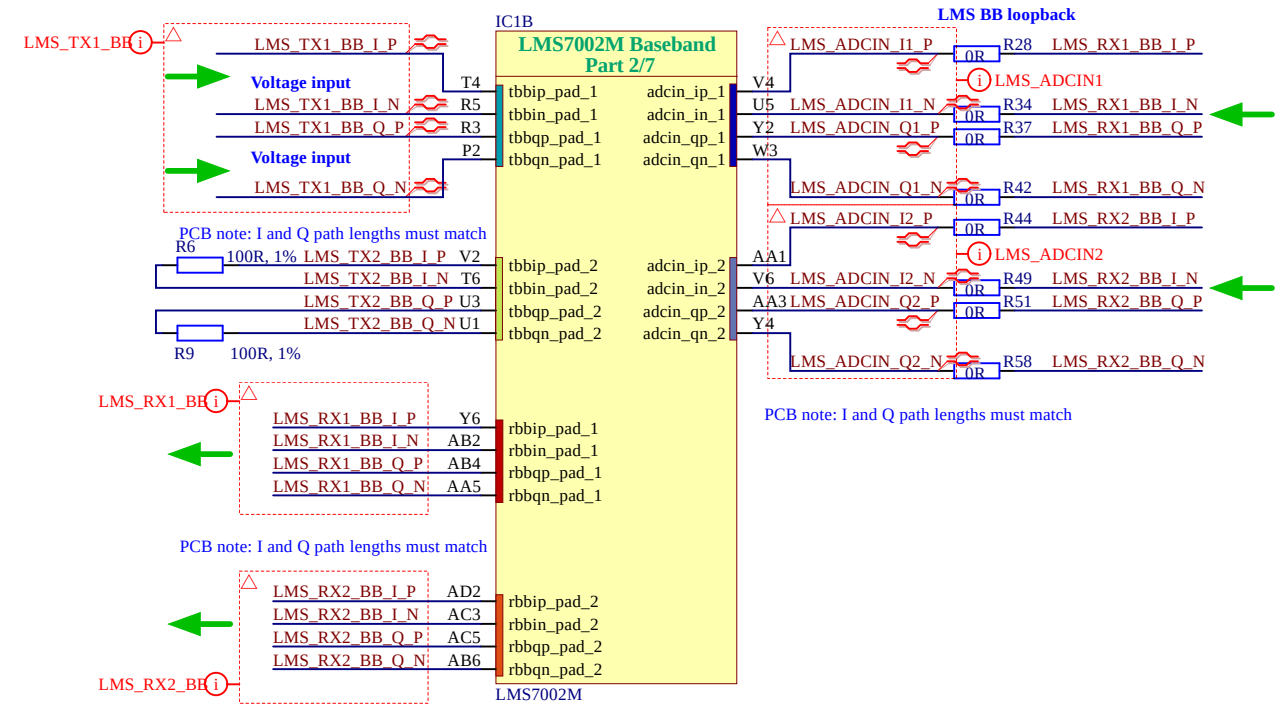
LMS7002M digital circuit



LMS EEPROM



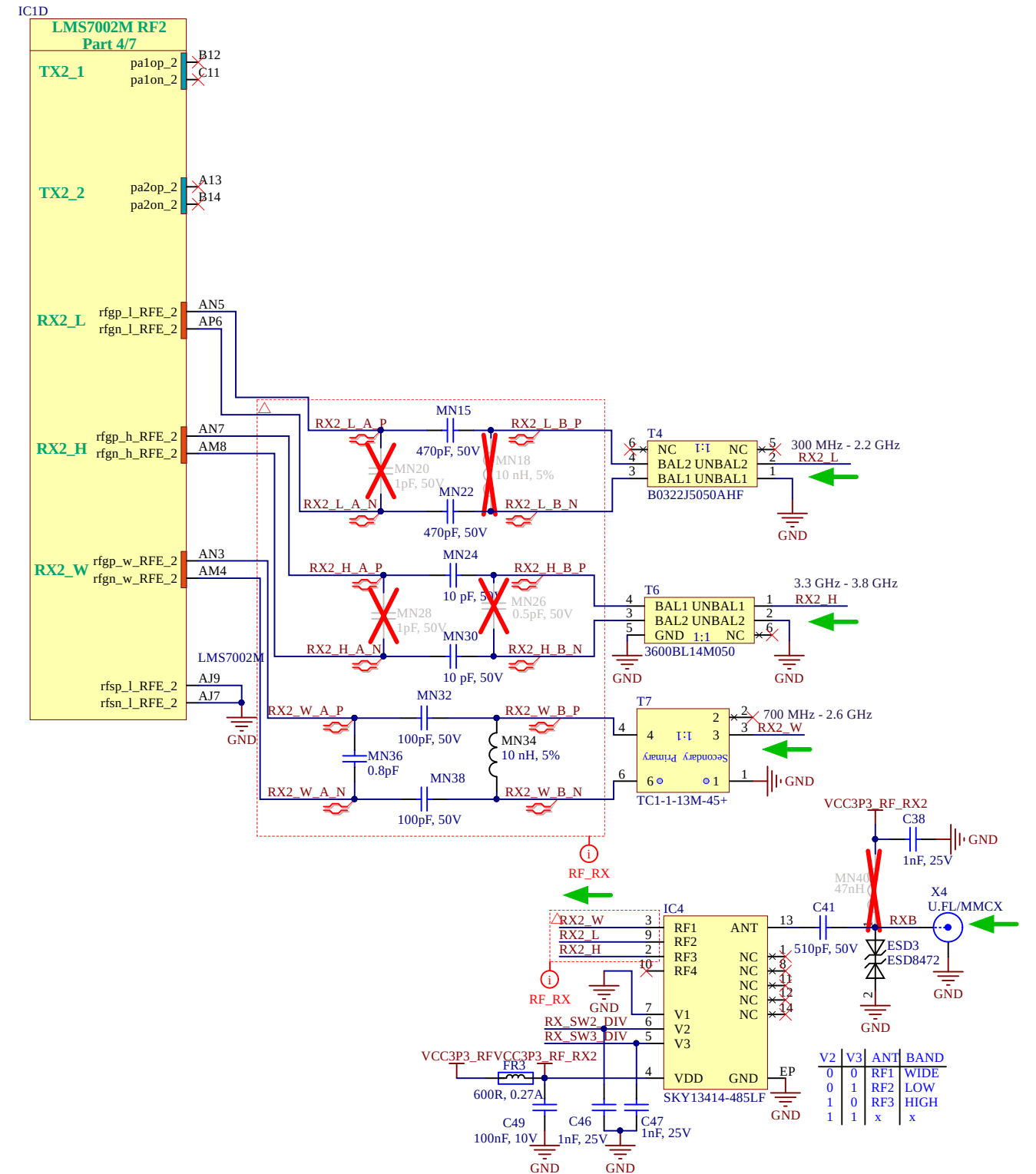
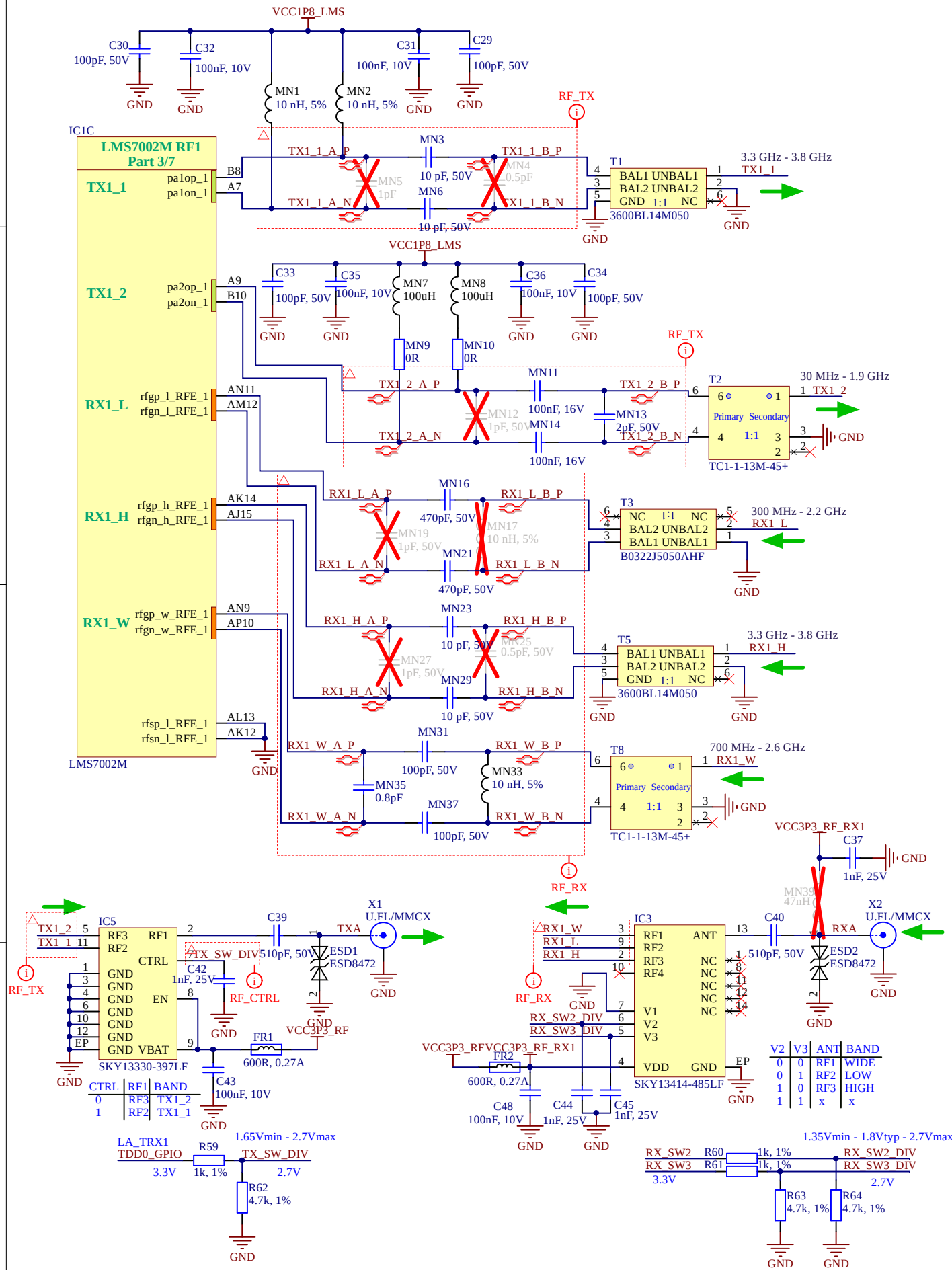
Baseband



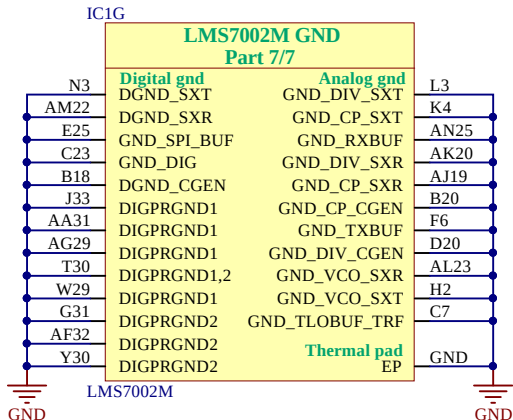
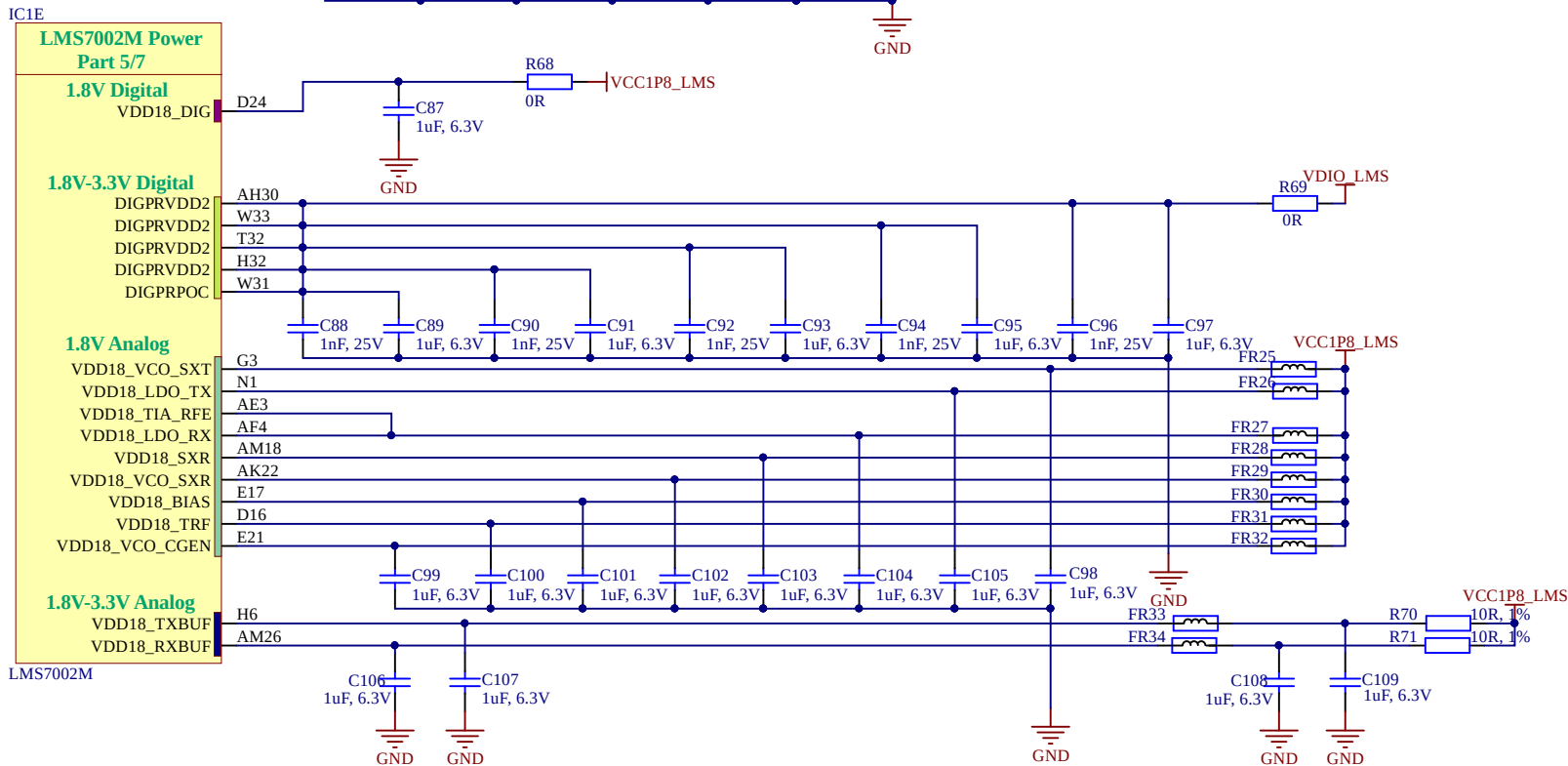
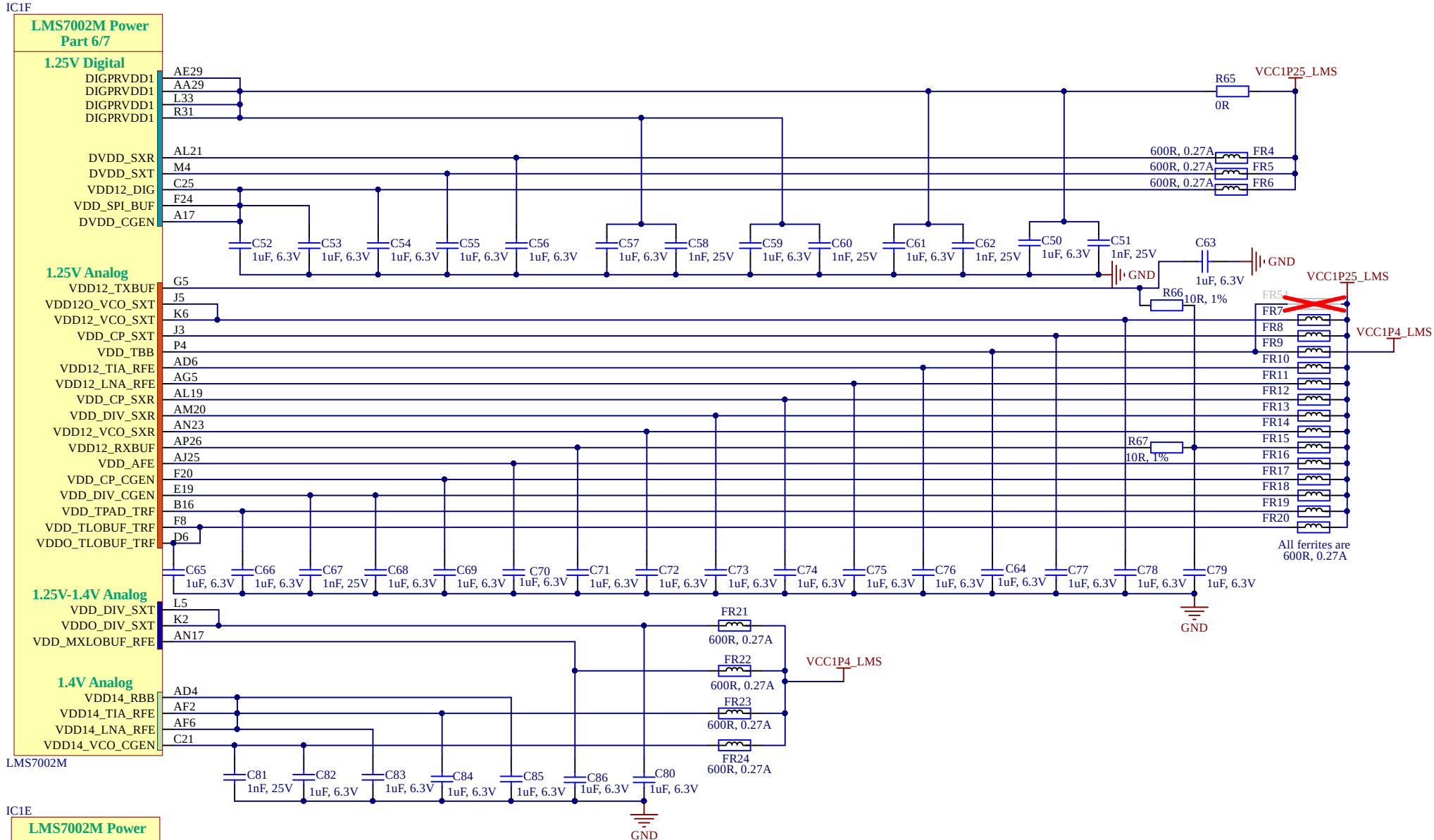
LMS7002M RF circuits

LMS RF Channel 1

LMS RF Channel 2



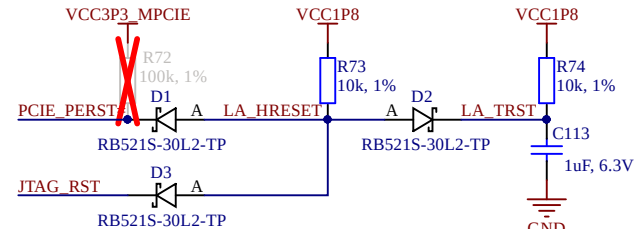
LMS7002M power supply circuit



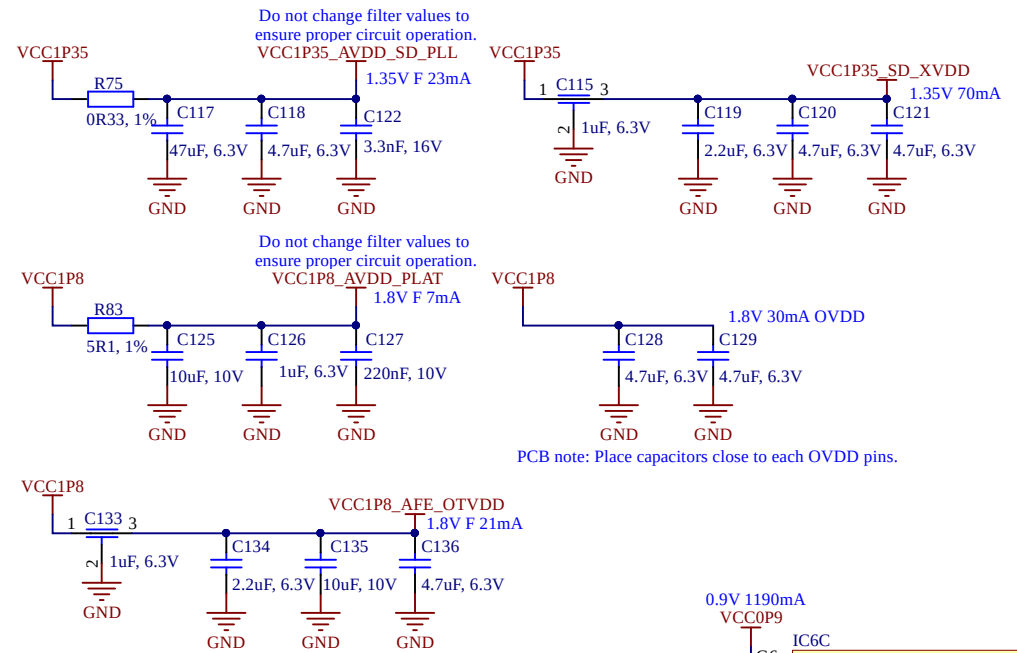
Project name: <i>LimeSDR-Micro_mPCIe_1v4.PrjPcb</i>				
Title: <i>LMS7002M power</i>		Lime Microsystems Surrey Tech Centre Guildford GU2 7YG Surrey United Kingdom		
Version: 1.4	Variant: Default			
Date: 6/16/2026	Time: 2:48:01 PM			Sheet 7 of 11
Edid: 07_J MS7002M PowerSchDoc	Size: A2			

Baseband Processor

Reset circuit

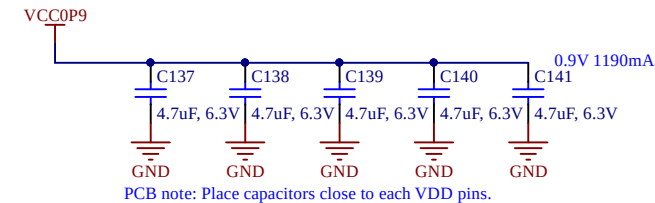


Power

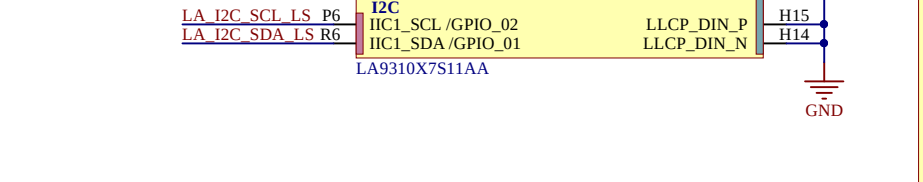
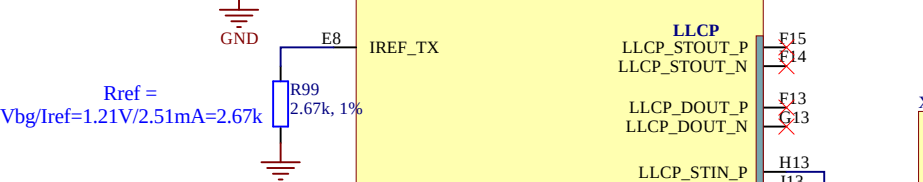
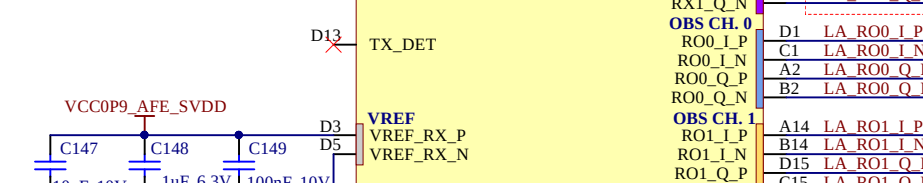
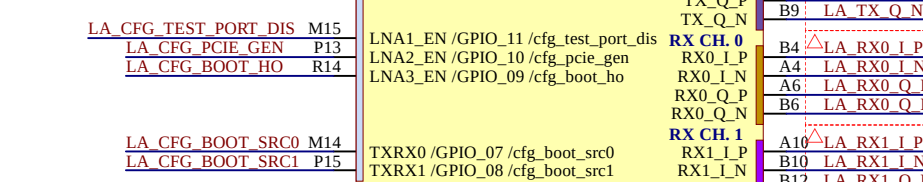
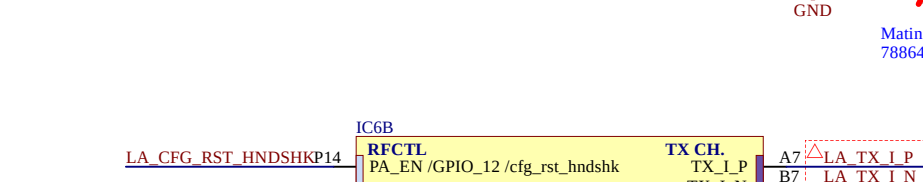
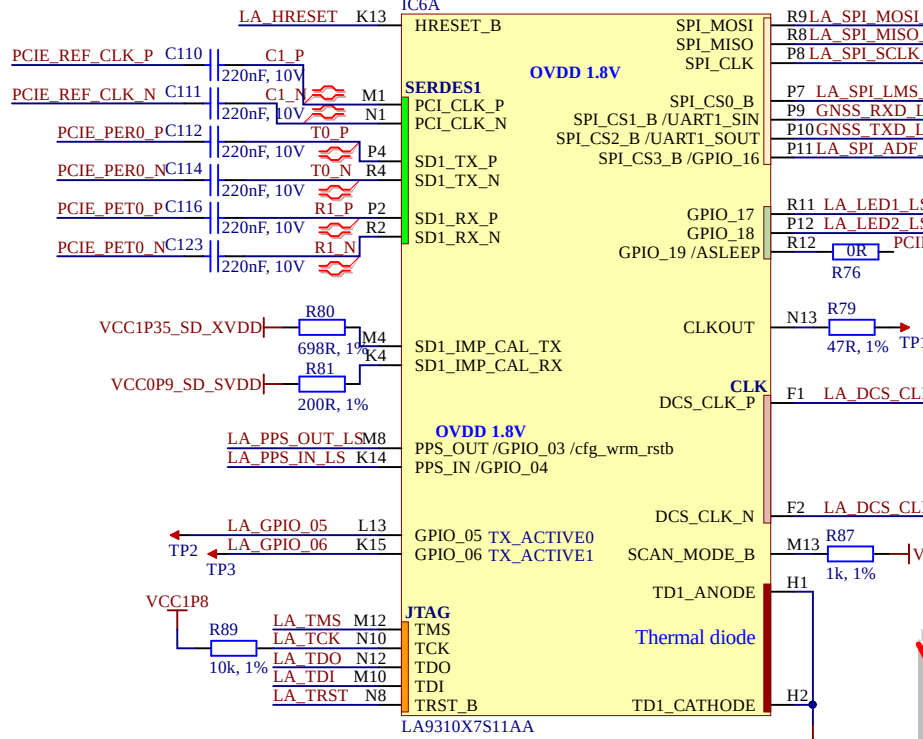
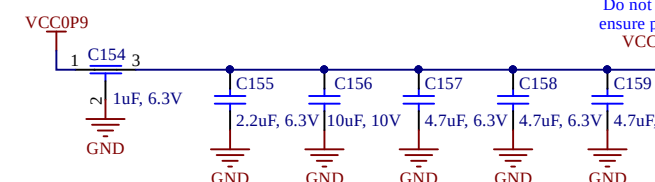
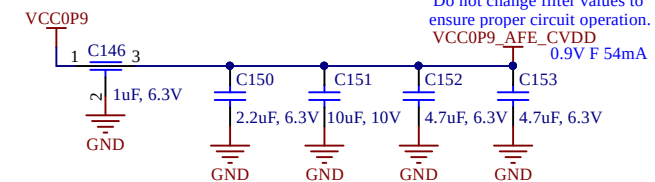
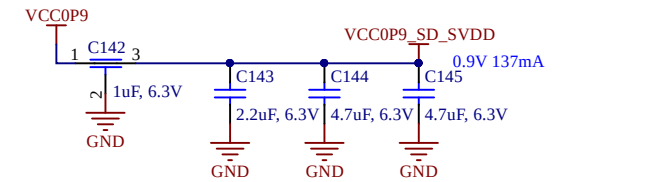


In LimeFEA mPCIe board pin 6 is connected to GND (same as for LimeSDR-XTRX). For compatibility with LimeFEA mPCIe board pin 6 (JTAG_RST) must be disconnected (0R NF).

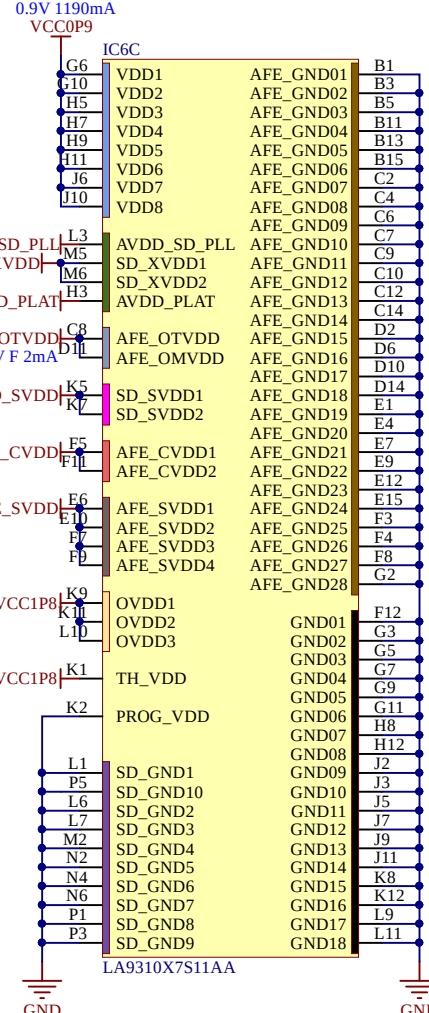
Mating to Molex 788641001 connector



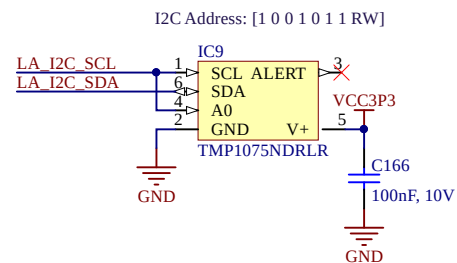
PCB note: Place capacitors close to each VDD pins.



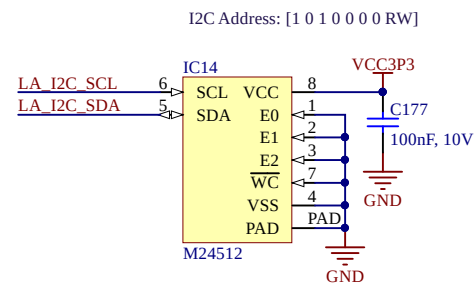
Enable internal common-mode voltage generation (0.45V) for all RX observation ADC inputs by setting ENACCPILING and ENVCMOUT bits in ADC Configuration Control Registers.



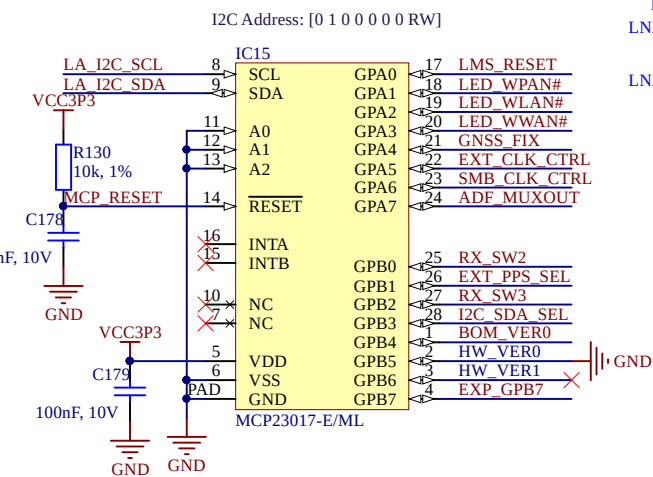
I2C Temperature sensor



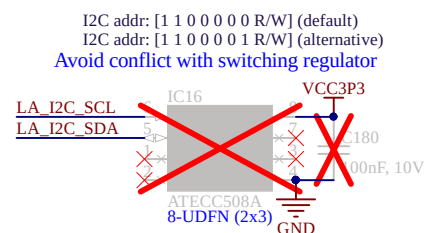
I2C EEPROM



I2C I/O expander

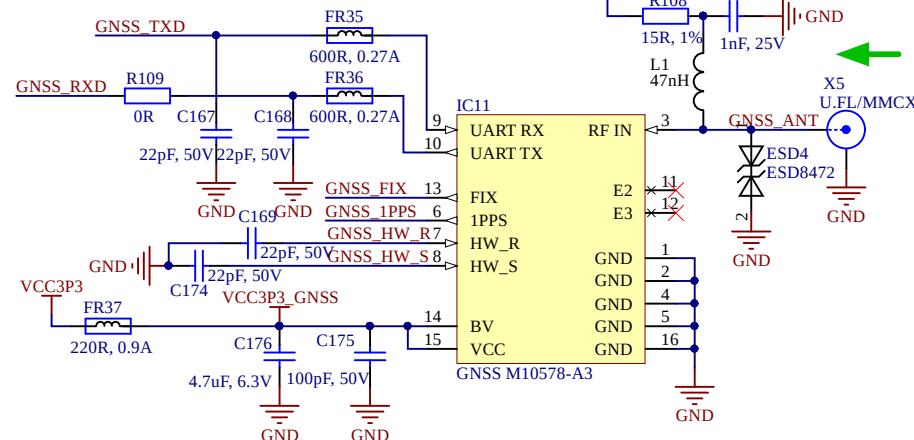


I2C secure key storage

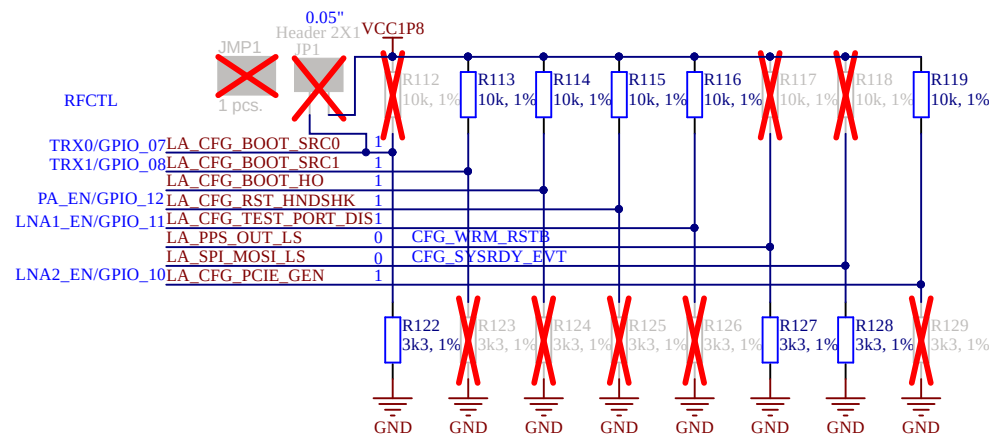


GNSS Receiver

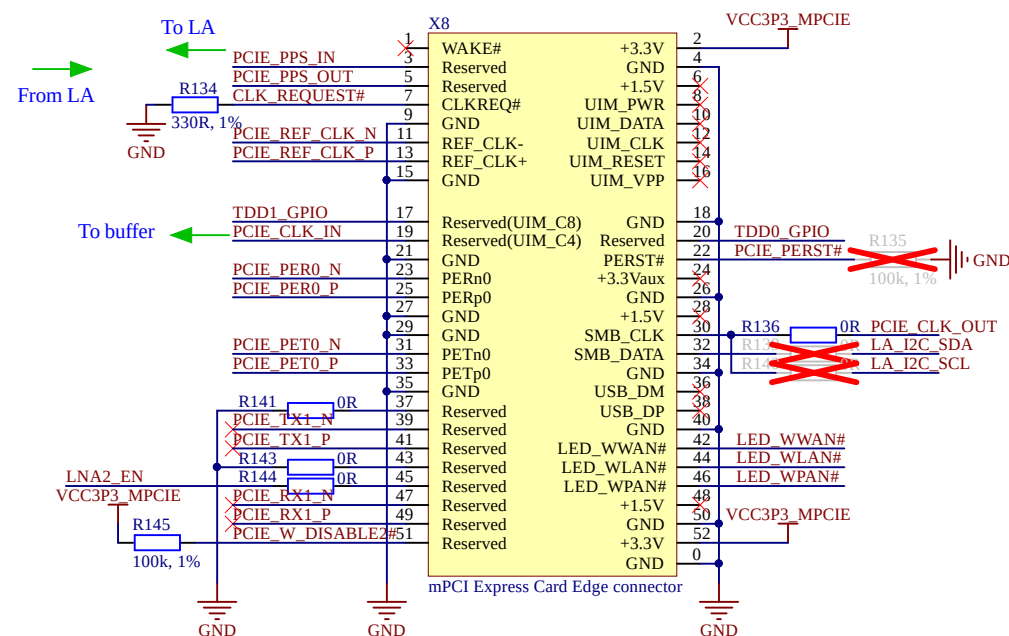
GND, GNSS_RXD and GNSS_TXD lines have 3 test pads near GNSS module (top PCB side) near edge



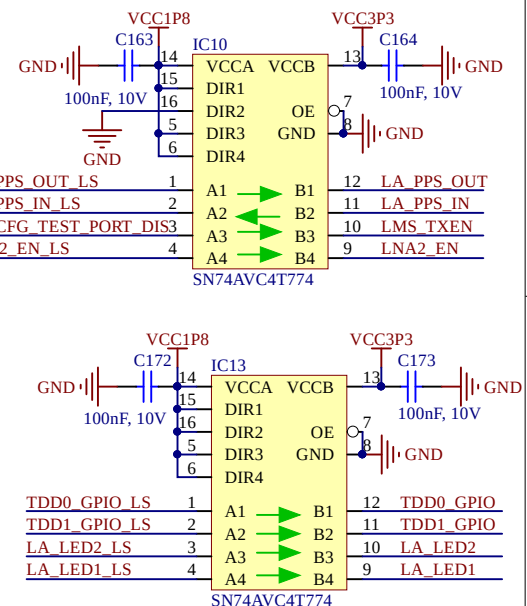
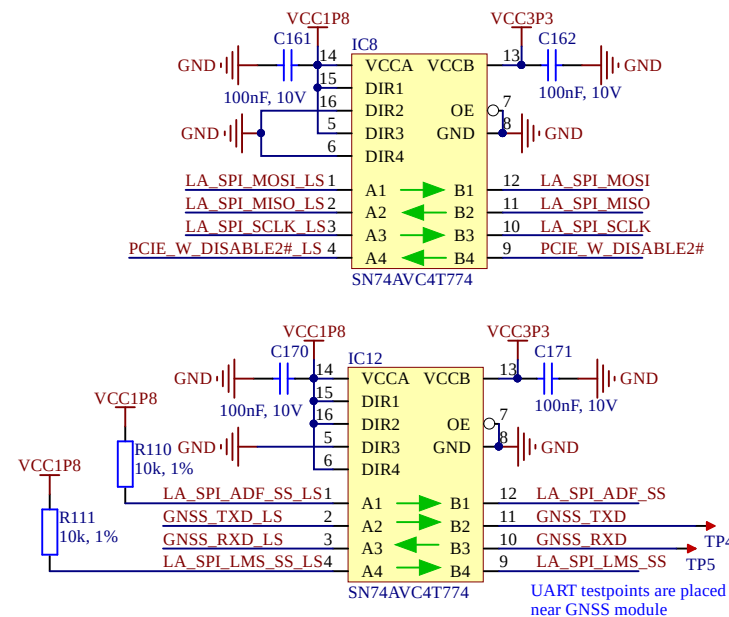
LA configuration



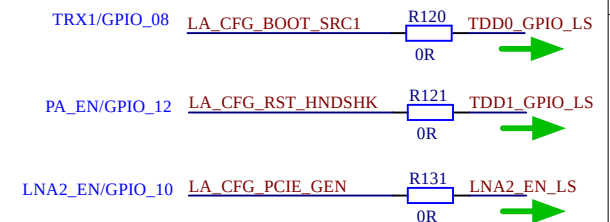
mPCIe



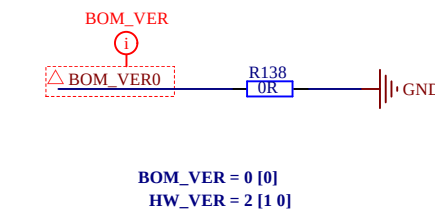
Level converters



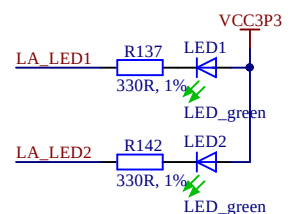
RFCTL GPIOs



HW_VER, BOM_VER

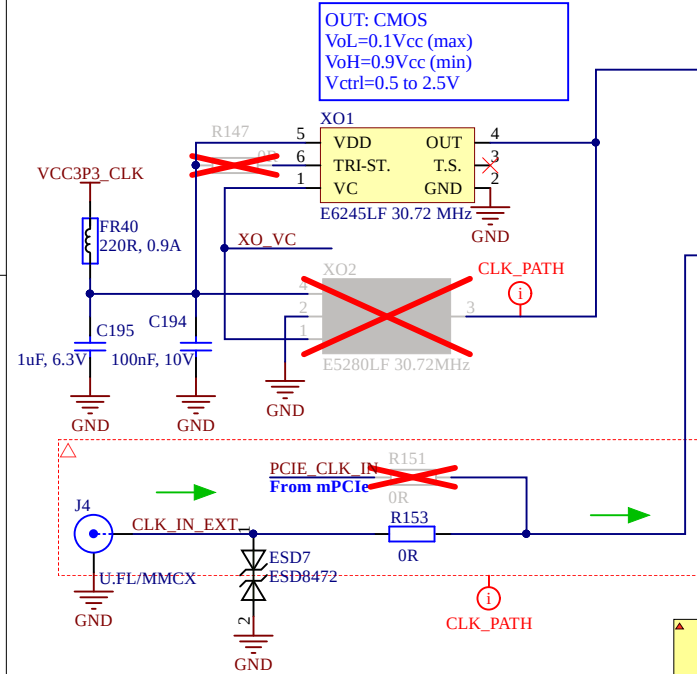


LA LEDs



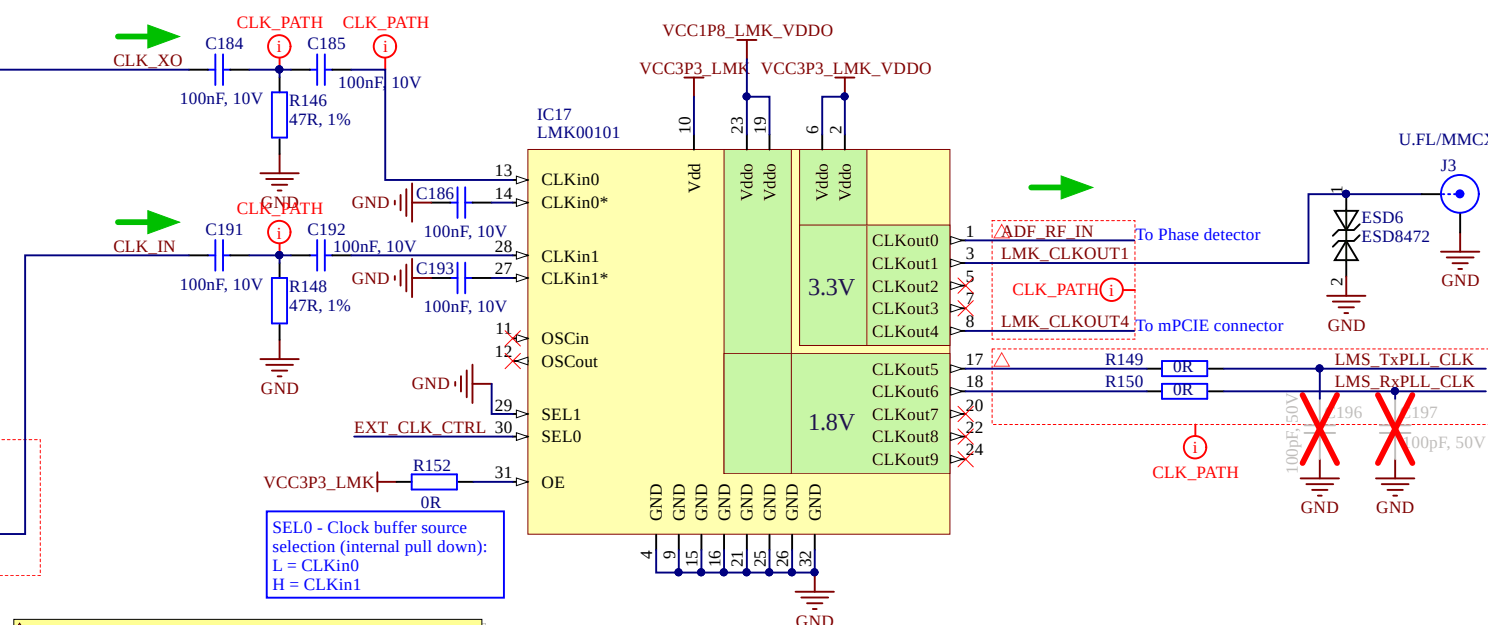
(VC)TCXO

OUT: CMOS
VoL=0.1V_{cc} (max)
VoH=0.9V_{cc} (min)
V_{ctrl}=0.5 to 2.5V



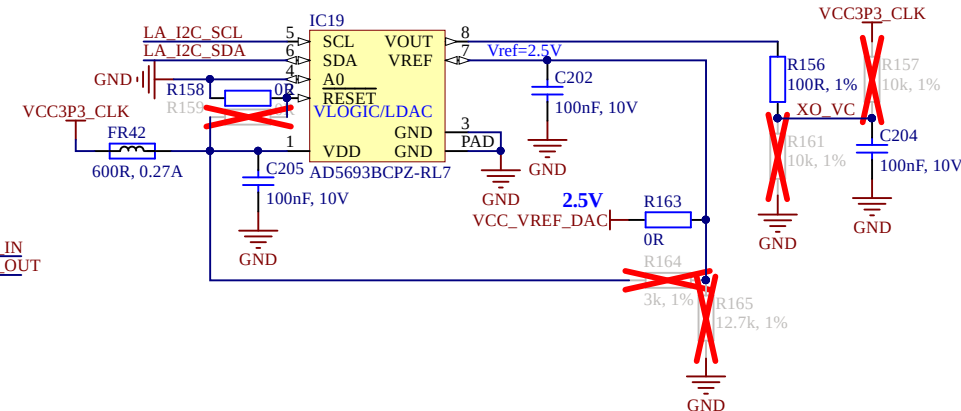
Clock

Clock buffer

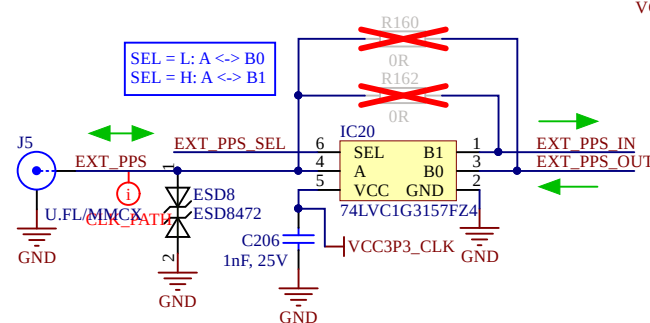


XO DAC

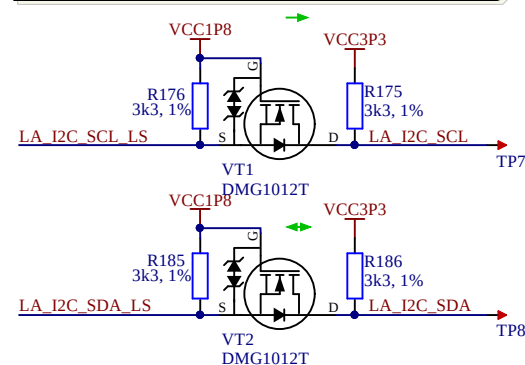
I2C Address: [1 0 0 1 1 0 0 RW]



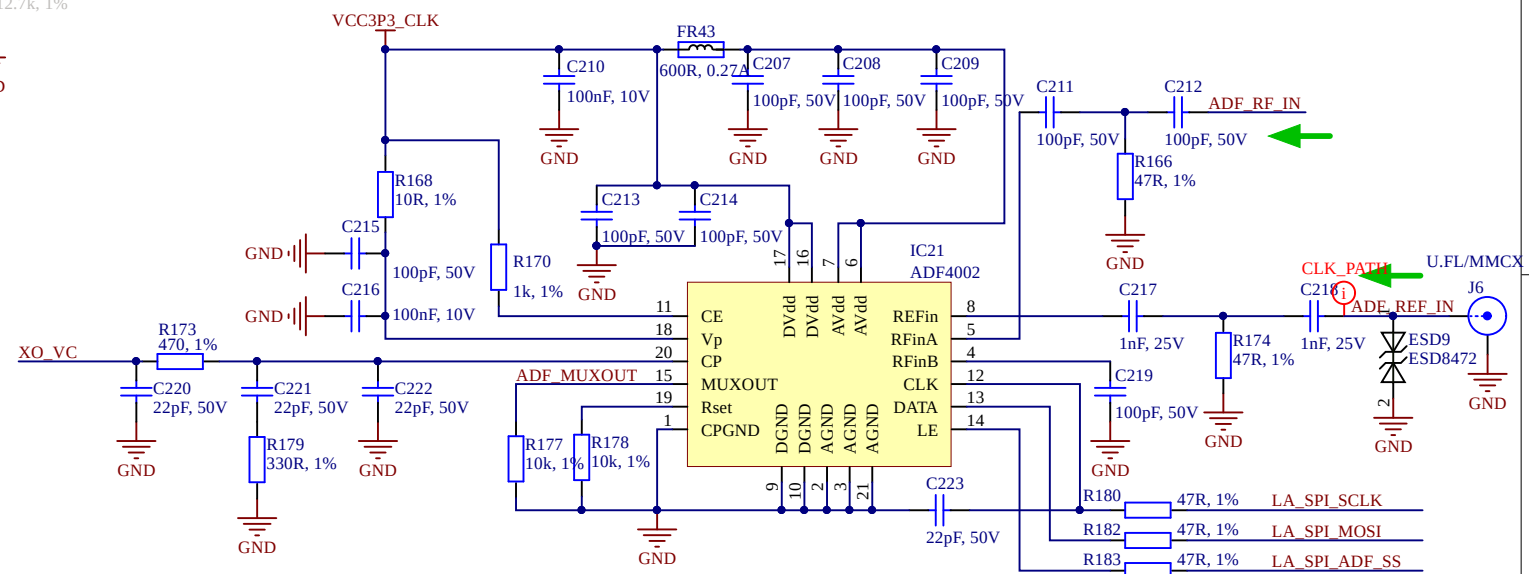
PPS selection



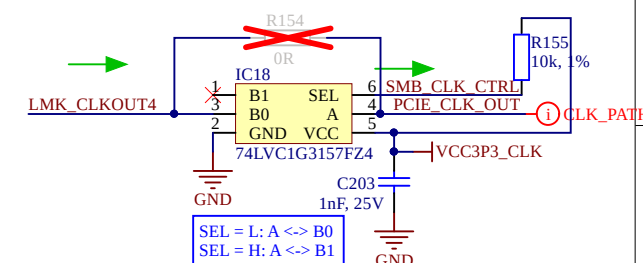
I2C level converters



Phase detector



mPCIe CLK OUT Switch



Project name: **LimeSDR-Micro_mPCIe_1v4.PriPcb**

Title: **Clock**

Version: 1.4

Variant: Default

Date: 6/16/2026

Time: 2:48:03 PM

Sheet 10 of 11

File: 10_Clock.SchDoc

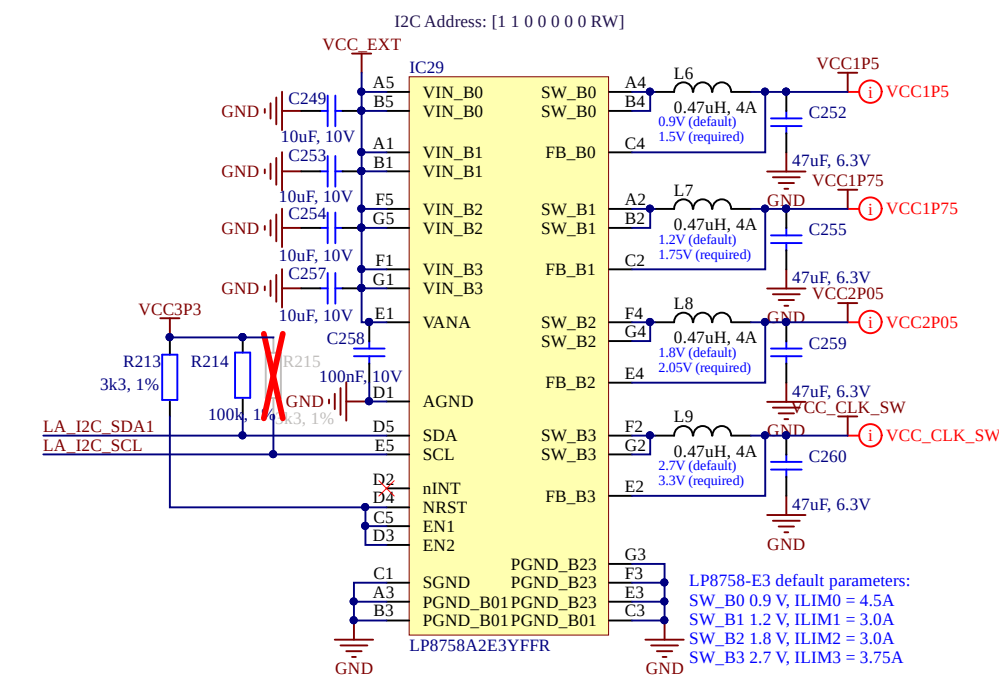
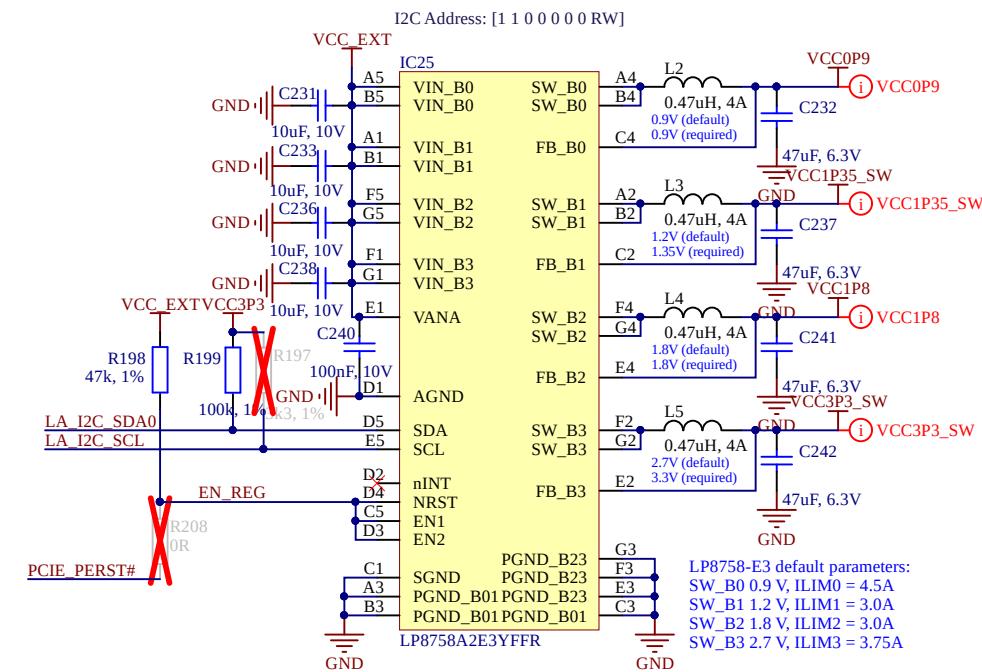
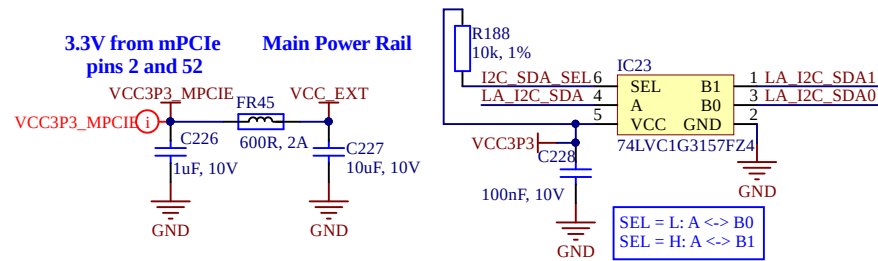
Size: A3

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Surrey
United Kingdom

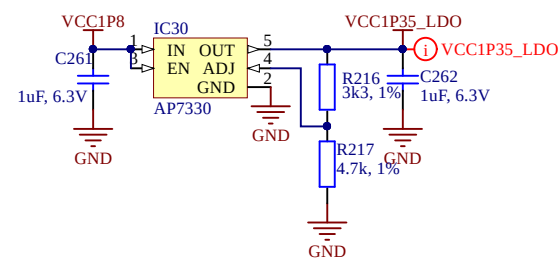
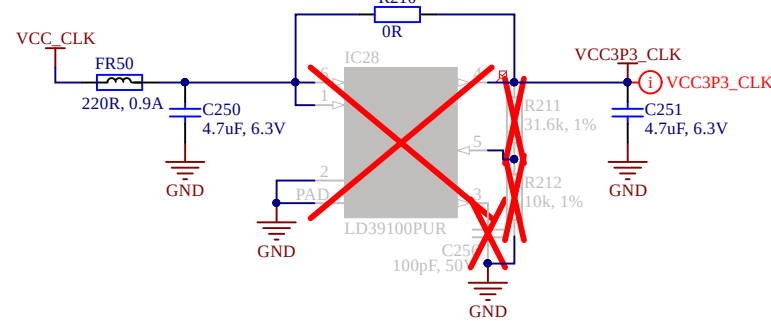
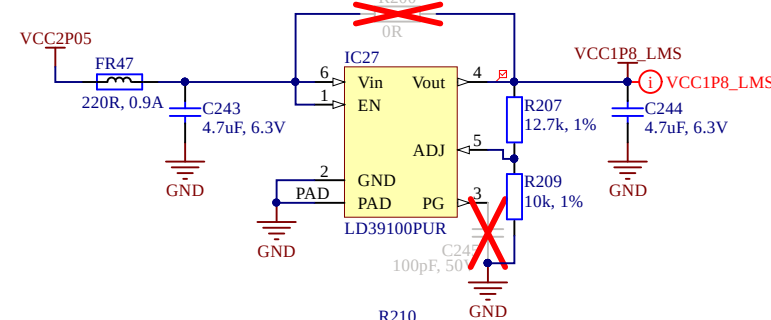
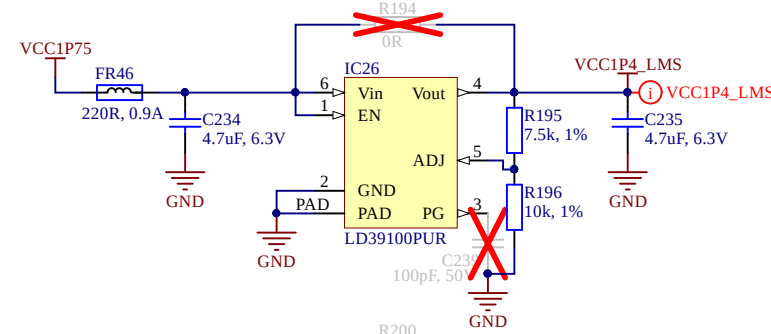
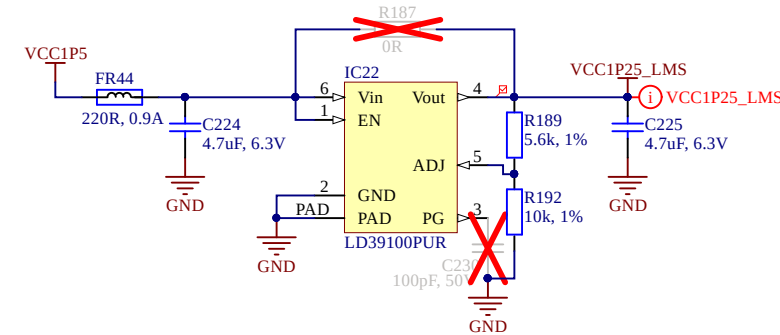


Board power circuits

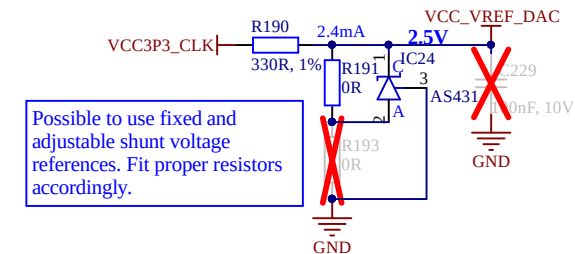
Switching regulators



Linear regulators

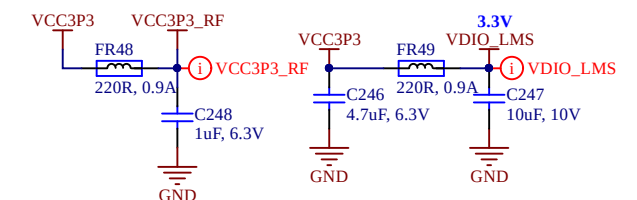
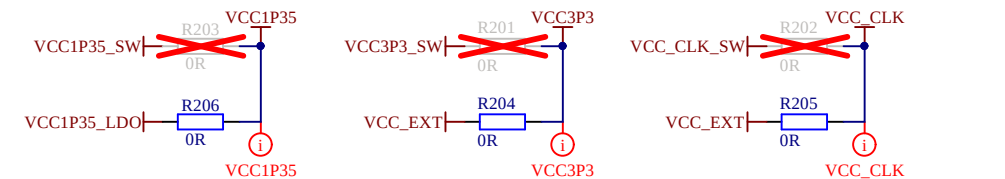


Voltage reference

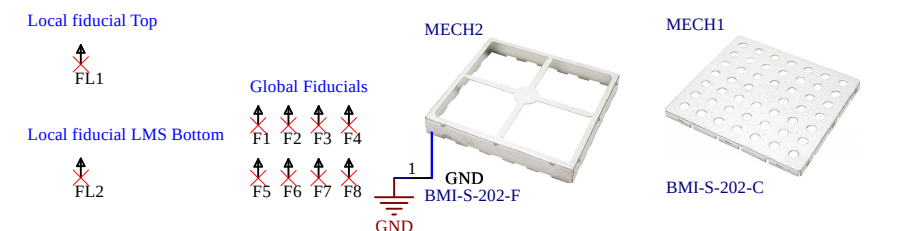


Possible to use fixed and adjustable shunt voltage references. Fit proper resistors accordingly.

Filters, voltage selection



Mechanical components



Project name: <i>LimeSDR-Micro_mPCIe_1v4.PrjPcb</i>			
Title: <i>Power</i>		<i>Lime Microsystems Surrey Tech Centre Guildford GU2 7YG Surrey United Kingdom</i>	
Version: <i>1.4</i>	Variant: <i>Default</i>		
Date: <i>6/16/2026</i>	Time: <i>2:48:04 PM</i>		
File: <i>11_Power.SchDoc</i>	Size: <i>A3</i>		